# Workshop 3: SQL

2026-04-08

## Contents



In this workshop, you will work in a team on a table with many other students around SQL. It is a collective discussion to find a solution to the given problem, step by step by exchange and comments. You need paper and pens or "draw.io" to draw the Entity-Relationship Diagram (ERD) and MariaDB with Docker to create the tables (see lecture 00 at the end). Please be positive, let all the people speak around the table, do not speak too loud, accept the comments for others, even negative comments can be treated as a way to improve your solution (always be positive). You can ask questions to me at any moment during the session. Välkomna, welcome, bienvenue…

## 1 Be positive! (5 min.)

Because, perhaps, you don't know each other around the table, take one minute per person to present yourself, explain who you are, where you come from, why you are taking this course/program and what you want to do in your future professional life. If everyone around the table is speaking Swedish, you can do it in Swedish, else, please go back to English.

## 2 Entity-Relationship Diagram (10 min.)

We consider the given Entity-Relationship Diagram for a library where Persons can borrow Books:

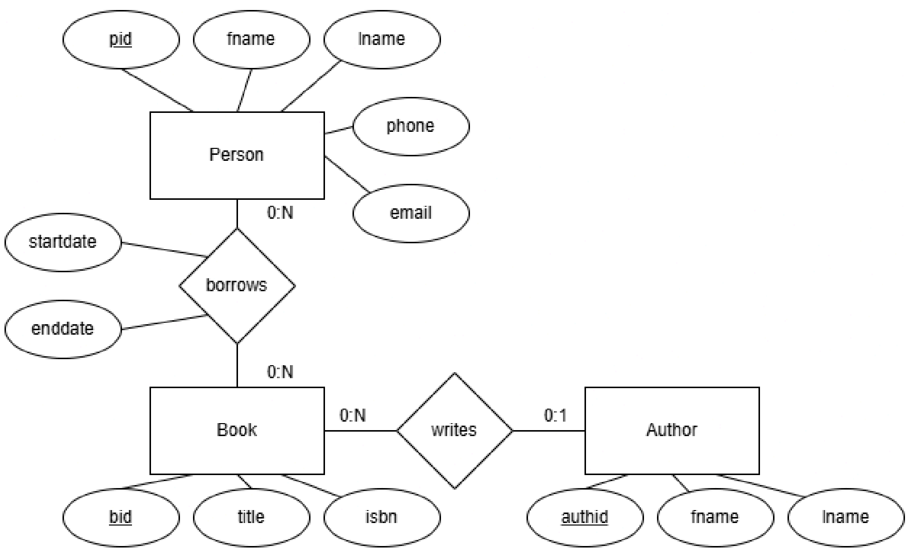


Figure 1: Persons borrowing books at the library.

Discuss with your colleagues around the table about the following questions (and please answer them together):

- Is this database in 3NF? Why?
- If you want the final table(s) to be in 3NF, what must you do?

- Transform this ERD into tables following the rules of the transformation from the conceptual to the logical model (see lecture 01 on page 29).

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Author(PK:authid, fname, lname)

Book(PK:

—

## 3 SQL DDL (10 min.)

- Install and run MariaDB with Docker (see lecture 00 at the end).
- Using the Data Definition Language (DDL) of MariaDB, create the necessary tables. Copy and paste your SQL code into a text file "workshop3.sql" so that you keep and save a version of your SQL commands.

## 4 SQL DML 1/2 (10 min.)

- Using MariaDB Data Manipulation Language (DML), insert data into the tables, populate the tables. You need three authors, five books, three persons and at least seven borrowings, three with startdate and enddate and two with no enddate (because the book is still borrowed). Copy and paste your SQL code into the text file "workshop3.sql" so that you keep and save a version of your SQL commands.
- Please, do not forget foreign keys when you insert data into the tables, otherwise, you will have to alter the table which is possible but more complicated.

## 5 SQL DML 2/2 (10 min.)

- Using SQL, write the queries to retrieve the desired information:
  - READ:
    - The list of all books.
    - The list of all authors.
    - The list of all persons.
    - The list of all borrowings.
    - The list of all borrowings before a certain date (startdate).
    - The list of all borrowings after a certain date (startdate).
    - The list of all borrowings still borrowed.
  - UPDATE:
    - Change the enddate of a borrowing.
    - Change the first name of an author.
    - Change the phone of a person.
  - DELETE:
    - Delete a borrowing.
    - Delete an author.
    - Delete a person.
- For each command, save the SQL code in the text file “workshop3.sql” so that you save a version of your SQL commands.

## 6 If you have finished…

In case you have finished the workshop very fast, you can work as a group on the lab assignment. I know you will also work with your partner on it, but it can help to start doing some parts of the SQL code. :) B+